YUSSUF MUBARAQ DAMILARE

Lagos Island, Lagos, Nigeria | Damilareoluwaseun54@gmail.com | +234 907 345 3591 / +234 814 403 8837
https://orcid.org/0009-0004-3105-7250 | *https://scholar.google.com/citations?user=5dHjh2EAAAAJ&hl=en*

RESEARCH PROFILE

Emerging translational medicine researcher bridging bench-level biochemistry and clinical application, with an M.Sc. thesis investigating oxidative stress biomarkers and histopathological outcomes in a diet-chemical-induced Type 2 diabetes rat model. Research experience spans molecular toxicology, medicinal-plant pharmacology, and nanomedicine drug delivery, complemented by hands-on clinical diagnostics and blood-transfusion laboratory practice. Seeking a Ph.D. position in Translational Medicine to extend this bench-to-bedside trajectory, with particular interest in metabolic disease models, oxidative-stress biomarker discovery, and nanotechnology-enabled drug delivery for chronic disease management.

RESEARCH INTERESTS

Metabolic Disease & Diabetes Research | Oxidative stress biomarker discovery | Medicinal plant pharmacology & phytochemistry | Nanomedicine & nanopharmaceutical drug delivery | Peptide-based drug design and biomaterial delivery systems | Translational & precision medicine | Phytochemical-drug interaction chemistry| Drug discovery and development processes

EDUCATION

M.Sc., Drug Research and Development Nov 2024 – Dec 2025
University of Lagos — Hybrid programme (online coursework with on-campus laboratory research and thesis)

- Thesis: "Effect of RAGV (1:1) extract on oxidative stress biomarkers and histopathological alterations in a Fructose–STZ induced Type 2 Diabetes rat model." Investigated antioxidant and tissue-protective effects of a combined plant-extract regimen in an induced-diabetes rodent model, generating biomarker and histopathology data relevant to translational diabetes therapeutics.

B.Sc. (Hons), Biochemistry Nov 2016 – Sep 2021
Lagos State University |

- Thesis: "Hypoglycemic and Antioxidant Activities of Carica papaya Leaf Extracts." Evaluated the hypoglycemic and antioxidant potential of C. papaya leaf extracts, contributing to the medicinal-plant evidence base later published as a peerreviewed article (see Publications).
- Relevant coursework: Experimental Biochemistry, Advanced Enzymology, Membrane Biochemistry, Microbial Biochemistry, Plant Biochemistry, Protein Chemistry, Clinical Biochemistry, Forensic Biochemistry, Genetic Engineering, Immunology, Biochemical Pharmacology.

PUBLICATIONS

First-Author Publications

1. Yussuf, M.D., Kokelu, A.N., Akinrodoye, A.O., Usoh, A.E., Kevin, E.A., Gomina, M., Usman, S., Ezeani, C.E., & Adimorah, M.I. (2025). Nanotechnology in Diabetes Management: Advances in Diagnostics, Drug Delivery, and Therapeutic Applications. *European Journal of Scientific Research and Reviews*, 2(2), 78–98.

doi:10.5455/EJSRR.20250108014459

2. Yussuf, M.D., Kokelu, A.N., Kelvin, E.A., Gomina, M., Usman, S., Nweze, V.C., Aluko, E.D., Umeh, I.E., & Bello, A.T.
   (2025). Medicinal Uses, Prevalence, and Mechanism of Action of Nigerian Plants with Psychoactive Properties. *Journal of Research in Complementary Medicine*, 2(2), 26–42. doi:10.5455/JRCM.20250113095500
3. Yussuf, M.D., Kokelu, A.N., Kolawole, H.O., Musa, A., Adisa, A.A.R., Ashimi, A.A., ... & Obasi, D.E. (2025). The Interplay of Apoptosis, Angiogenesis, and Metastasis in Cancer Pathogenesis. *Journal of Cancer Research and Reviews*, 2(2), 37.
4. Yussuf, M.D., Ikedionu, C.A., Ocheinehi, A.J., Ugwuanyi, I., Olaoye, E.B., & Olawuyi, M.D. (2025). Biodegradable Peptide Hydrogels for Precision Drug Release: Advances and Future Perspectives. *European Journal of Scientific Research and Reviews*, 2(3), 130.
5. Yussuf, M.D., Ugwuanyi, I., Oghenetejiri, O.U., Awegbe, C.U., Akinyanju, B.J., Ifebuzor, C.B., & Olaoye, E.B. (2025). Peptide Hydrogels as Biocompatible Carriers for Sustained Antibiotic Release Against Multidrug-Resistant Infections. *European Journal of Scientific Research and Reviews*, 2(3), 154.
6. Yussuf, M.D., Edozie, K.S., Eze, C.Y., Okei, N.C., Obasi, D.E., & Akinrodoye, A.O. (2025). Drosophila melanogaster and its Genetic Tools as a Model in Neurological Disorder Research. *European Journal of Scientific Research and Reviews*.
7. Yussuf, M.D., Umeh, I.E., Kolawole, H.O., Adisa, A.A.R., Ashimi, A.A., Oladosu, M.A., Obasi, D.E., Olumide, M.K., & Ayo-ige, A.B. (2025). The Role of Hormones in the Onset and Progression of Endocrine and Gynaecological Cancers.
   International Journal of Advanced Biology and Biomedical Research, 13(5), 573–603. doi:10.48309/IJABBR.2025.2056347.1585
8. Yussuf, M.D., Samuel, A.O., Dayo, U., Abdul, S.O., Stanley, O.C., & Okunade, A. (2024). Lipophilic Vitamins (D and E) and Flavonoids (Quercetin and Naringin) Attenuate the Effects of Dichlorvos Toxicity on Catalase Activity. *International Journal of Advanced Biological and Biomedical Research*, 12, 67.

Co-Authored Publications

9. Izah, C.N., Jonathan, F., Puorizaa, T., Aloh, M.T., Eneje, I.L., Okoye, A.I., ... & Yussuf, M.D. (2025). Statistical Evaluation of Structure-Activity Relationships (SAR) in Antimalarial Drug Development: A Medicinal Chemistry Perspective. *European Journal of Scientific Research and Reviews*, 3(2), 128.
10. Kokelu, A.N., Fadipe, D.P., Umeh, I.E., Lawal, F.G., Chukwudum, E.D., Ikedionu, O.I., & Yussuf, M.D. (2025). Advances in Translational Immunotherapy for Autoimmune Diseases: from Target Discovery to Patient-Specific Therapies. *European Journal of Scientific Research and Reviews*, 3(4), 240.
11. Ojukwu, U.C., Arthur, C., Jonathan, W., Ehigiegba, O.J., Yussuf, M.D., & Olawuyi, M.D. (2025). Green Chemistry in Pharmaceutical Synthesis: Implications for Sustainable Public Health. *European Journal of Scientific Research and Reviews*, 3(1), 1.
12. Kokelu, A.N., Ologunja, O.J., Anyawu, T.I., Shittu, T.H., Kuboye, T.B., Ezem, F.C., ... & Yussuf, M.D. (2025). Precision Medicine in Chronic Disease Prevention and Management. *Journal of Medical Research and Reviews*, 4(1), 28.
13. Omiyale, O.C., Ogunjobi, T.T., Babalola, G.J., Olofin, O.O., Akib, M.O., Yussuf, M., Shodipe, O.O., Ogbuefi, C.F.C., Agyei-acheampong, C.A., Oparah, C.D., Kolapo, E.E., & Olowe, E.N. (2025). A Review on Inflammation and Cardiovascular Diseases: What is New? *European Journal of Scientific Research and Reviews*, 2(1), 23.
    doi:10.5455/EJSRR.20240602093947
14. Omiyale, O.C., Ogunniran, B.I., Ezeh, M.O., Yussuf, M.D., Oparah, C.D., & Isshak, F. (2024). Hypoglycemic and
    Antioxidant Activities of Carica papaya Leaves. *International Journal of Advanced Biological and Biomedical Research*, 12, 141–154. *(companion publication to B.Sc. thesis)*

15. Omiyale, O.C., Yussuf, M.D., Kehinde, E.O., & Olofin, O.O. (2024). Effects of Jicama (Pachyrhizus erosus)-Based Oral
Rehydration Solution on Intestinal Alkaline Phosphatase in Diarrhoeagenic Rats. *International Journal of Advanced*
Biological and Biomedical Research, 12(2), 16–25. doi:10.48309/IJABBR.2024.2017216.1472
16. Charles, O.O., Damilare, Y.M., Damian, O.C., Eriuroro, J.P., Precious, I., Dolapo, I.A., ... & Hannah, A. (2023). Phytochemical Profile and Antioxidant Properties of Extracts from Sargassum flutans. *International Journal of Advanced Biological and Biomedical Research*, 11(4), 261–276.

CONFERENCE PRESENTATIONS & POSTERS

- Best Poster Presentation Award — International Conference of Biochemistry: "Effects of Momordica charantia (Bitter Lemon) on Ischemic Diabetic Myocardium"
- Poster, 3rd International Conference on Public Health in Africa (CPHIA 2023): "Effects of Garlic Extract on TNF-α Expression and Oxidative Stress Status in the Kidney of Rats with STZ + Nicotinamide-Induced Diabetes"
- "Anti-Diabetic Activity of Stigmasterol from Soybean Oil by Targeting the GLUT-4 Glucose Transporter"
- "Silymarin Ameliorates Diabetic Cardiomyopathy via Inhibiting TGF-β1/SMAD Signalling"

RESEARCH & CLINICAL LABORATORY EXPERIENCE

Laboratory Analyst May 2024 – Present
Lagos Island General Hospital

- Perform diagnostic testing across hematology, chemistry, and microbiology, applying quality-control protocols and calibrated instrumentation (analyzers, centrifuges, microscopes) consistent with translational research standards for sample integrity.
- Document and communicate findings to clinicians, contributing to diagnosis and treatment planning.
- Maintain infection-control and biohazard protocols within a regulated clinical laboratory environment.

Laboratory Analyst (Intern) Jan 2023 – Mar 2024
Lagos State Blood Transfusion Centre

- Conducted blood cross-matching, antibody screening, and ABO/Rhesus typing; performed serological testing for human retroviruses and viral hepatitis B & C.
- Maintained laboratory records and specimen logs in accordance with blood-bank protocols.

TEACHING EXPERIENCE

Science Teacher (Chemistry & Biology) Oct 2021 – Nov 2022
Fumsville College

- Designed and delivered Chemistry and Biology lessons and practical experiments aligned with curriculum standards; maintained laboratory equipment and safety compliance.
- Recipient of the Excellence in Teaching Award, Ministry of Education, and the Fumsville College Excellence Award.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Precision Medicine — University of Geneva (Coursera, 2024)
- Statistics and Data Analysis in Public Health — Imperial College London (Coursera, 2024)
- Understanding Clinical Research: Behind the Statistics — University of Cape Town (Coursera, 2024)

- Introduction to Drug Discovery and Development — Udemy (2024)
- Financial Modelling Workshop — Udemy (2024)
- Basic Life Support (BLS) Certification — Postgraduate Institute of Medicine (2024)
- Use of Epidemic Intelligence Systems for Pandemic Preparedness — OpenWHO.org, WHO Health Emergencies Programme (2024)
- Monitoring and Evaluation of National Action Plans on Antimicrobial Resistance — OpenWHO.org, WHO Health Emergencies Programme (2024)
- Data Analytics, UI/UX and Machine Learning — Udemy (2024)
- Complete Python and Machine Learning in Financial Analysis — Udemy (2024)
- Lifestyle Management and Pharmacological Treatment Options for People with Diabetes and Chronic Kidney Disease – American Diabetes Association (2026)
- Engage to Empower; Screening, Diagnosis and Staging of Chronic Kidney Disease—American Diabetes Association (2026)

AWARDS & HONORS

- Outstanding Researcher Award — International Journal of Advanced Biological and Biomedical Research (2023)
- Best Poster Presentation Award — International Conference of Biochemistry
- Excellence in Teaching Award — Ministry of Education
- Fumsville College Excellence Award

PROFESSIONAL AFFILIATIONS

- American Society for Microbiology
- Nigerian Society of Biochemistry and Molecular Biology
- Genetics Society of America
- British Society for Antimicrobial Chemotherapy

TECHNICAL & RESEARCH SKILLS

- Laboratory techniques: GC-MS, spectrophotometry, PCR, Western blotting, in vivo animal modelling, histopathology • Statistical analysis and research methodology
- Clinical diagnostics: hematology, serology, blood banking (cross-matching, antibody screening, ABO/Rh typing) • Software: Microsoft Word, Excel, PowerPoint

REFERENCES

Available upon request